

Industrial Reliability and Lubrication Best Practices Feasibility Study

Anonymized Engineering Economics Academic Case Study

Course: DS-301 / IE0361 Engineering Economics

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Executive Summary

This study evaluates the economic and operational value of changing a manufacturing site's reactive lubrication practices to a structured Lubrication Best Practices (LBP) program. The case is anonymized for confidentiality. The technical method, assumptions, and financial results are retained.

The baseline cost of lubricant consumption, hydraulic-component repair, and unplanned downtime was estimated at approximately 96,250 JOD per year. Two alternatives were evaluated over five years: a basic program using training, desiccant breathers, and filtered oil stations; and an advanced program using higher-specification breathers, filtration, dispensing, and condition-based monitoring.

Decision

Both alternatives are economically viable at a 12 percent minimum attractive rate of return (MARR). The advanced package is recommended where long-term reliability, contamination control, and predictable lifecycle cost are more important than the lowest initial investment.

Case Context

The studied manufacturing site operates hydraulic equipment continuously. Frequent valve and pump failures were linked to contamination, inefficient oil-change intervals, and inconsistent oil handling. The study therefore considered practices that reduce contamination, extend component life, and replace time-based maintenance with condition-informed decisions.

Objectives

- Compare basic and advanced LBP investment packages.
- Apply present worth, annual worth, future worth, internal rate of return, payback period, and benefit-cost analysis.
- Test sensitivity to maintenance-cost escalation, inflation, and MARR changes.
- Recommend the economically stronger option while noting operational trade-offs.

Financial Inputs and Alternatives

Metric	Basic LBP	Advanced LBP
Initial investment	49,615 JOD	86,863 JOD
Annual savings	50,959 JOD	79,089 JOD
Present worth at 12%	134,081 JOD	198,235 JOD
Internal rate of return	About 99%	About 87%

Metric	Basic LBP	Advanced LBP
Payback period	1.10 years	1.26 years
Benefit-cost ratio	3.70	3.28

The baseline includes approximately 84,000 JOD per year for hydraulic repair and 12,250 JOD per year for lubricant. The analysis assumes a five-year study period, constant annual savings, and a 12 percent MARR. Breathers require periodic replacement, while oil stations and filtration equipment have a service life exceeding the analysis horizon.

Method

Present worth discounts all expected cash flows to the beginning of the study. Annual worth converts that value into an equivalent annual amount. Future worth expresses the terminal value, while the internal rate of return identifies the discount rate at which net present value becomes zero. Payback period measures investment recovery time and the benefit-cost ratio compares discounted benefits with discounted cost.

Risk and Sensitivity Review

The sensitivity analysis tested maintenance-cost increases of 20 percent and 50 percent. Both packages stayed economically viable, but the advanced package retained a stronger and more stable financial profile under the severe case. Its higher initial investment is offset by more durable contamination-control measures and lower exposure to lifecycle variability.

Scenario	Basic LBP	Advanced LBP
20% maintenance-cost increase	IRR falls to about 80%; B/C about 3.04	Payback remains near the base case; B/C about 3.38
50% maintenance-cost increase	IRR about 49%; payback extends to years 3-4	Payback remains near 1.26 years; B/C remains above 3

Inflation and MARR scenarios show the expected reduction in present worth, benefit-cost ratio, and equivalent annual worth as the nominal discount rate rises. The comparative decision remains unchanged because the advanced package maintains higher overall present worth and better resilience under the tested adverse cases.

Recommendation

Select the advanced LBP package when the objective is the largest long-term reduction in hydraulic-system risk and operating cost. Select the basic package when capital availability and the fastest payback are the primary constraints. In either case, condition-based oil monitoring, controlled dispensing, filtration, and training are central to sustaining the predicted benefits.

Confidentiality Note

This public version removes the external site's name and operating-unit identifiers. No confidential company information is intended to be disclosed.